KPI Overview

Workforce Scheduling Optimization — SmarTask / Sisqual

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This document lists all Key Performance Indicators used to evaluate workforce schedules, grouped by category. Each KPI is described by its name and what it measures or represents.

Contents

Category 1 — Hard Constraint Alarms

These KPIs verify whether the schedule respects legally and contractually binding rules. Each one reports the number of violations found; the target is always zero.

ID	Name and description
ALM-1	**Duplicate assignments per day** Checks that no employee has been assigned more than one shift on the same day.
ALM-2	**Uncovered slots without a skill** Checks that every working time slot assigned to an employee has a skill associated with it (ALM, EG, or CAJ). No slot should be left without a task.
ALM-3	**Maximum consecutive working days** Checks that no employee works more than 5 days in a row without a rest day.
ALM-4	**Minimum rest between shifts** Checks that every employee has at least 11 hours of rest between the end of one shift and the start of the next.
ALM-5	**Work on unavailable days** Checks that no employee is assigned a shift on a day marked as unavailable (fixed day off, vacation, sick leave, or general absence).
ALM-6	**Sunday rest per month** Checks that every employee has at least one Sunday off during the scheduling month.
ALM-7	**Weekly contracted hours** Checks that the total hours assigned to each employee each week match their contracted hours.

Category 2 — Alarm Priority Hierarchy

The system must follow a predefined priority order to fulfill the minimum requirements. These KPIs measure how many time slots each priority level failed to meet its minimum staffing requirement.

Priority	Task and description
1	**Warehouse supervisor (level $\geq$ 1)**
2	**Management supervisor (level 1)**
3	**Cashier (level 1)**
4	**Management supervisor (level 2)**
5	**Cashier (level 2)**
6	**Management supervisor (level 3)**
7	**Management supervisor (level $\geq$ 4)**
8	**Cashier (level 3)**
9	**General employees (any level)**

Category 3 — Performance and Objective Function Metrics

These KPIs measure the quality of the schedule in terms of how well it satisfies staffing demand. They are directly linked to the objective functions of the optimisation model.

ID	Name and description
PERF-1	**Total staffing shortage** Total number of worker-slots where the minimum staffing requirement was not met, summed across all days, time slots, and skills.
PERF-2	**Severity index** A weighted version of the total shortage that gives more importance to failures at higher-priority alarm levels. A shortage at priority 1 counts more than one at priority 9.
PERF-3	**Ideal staffing shortage** Total number of workers missing across all time slots to reach the ideal staffing level.
PERF-4	**Estimated staffing shortage** Total number of workers missing across all time slots to reach the estimated staffing level.
PERF-5	**Shortage by skill** Breakdown of the total shortage (PERF-1) by skill category: ALM (warehouse), EG (management), and CAJ (cashier).
PERF-6	**Number of slots with at least one failure** Count of individual time slots where at least one skill requirement was not met. Measures the breadth of coverage failures.
PERF-7	**Maximum shortage in a single slot** The largest shortage recorded in any single time slot. Identifies the worst-case bottleneck in the schedule.
PERF-8	**Mean shortage per demanded slot** Average shortage per time slot that had a staffing requirement. Normalises the total shortage by the number of demanded positions, making it comparable across instances of different sizes.

Category 4 — Per-Employee KPIs

These KPIs provide a breakdown of schedule quality at the individual employee level. They can be filtered by employee and by day.

ID	Name and description
EMP-1	**Total worked hours** Total number of hours assigned to the employee across the entire scheduling period.
EMP-2	**Contracted hours utilisation rate** Percentage of contracted hours that were actually assigned. Values above 100% indicate overload; values significantly below 100% indicate idle capacity.
EMP-3	**Number of working days** Total number of days on which the employee was assigned a shift.
EMP-4	**Maximum consecutive working days** The longest streak of consecutive working days for the employee. Must not exceed 5 (see ALM-3).
EMP-5	**Skill usage percentage** Breakdown of the employee's working time by skill (ALM, EG, CAJ). Shows how each employee's time is distributed across tasks. Useful for detecting whether multi-skilled employees are being used appropriately.
EMP-6	**Daily assignment segments** Number of distinct skill blocks per day for each employee. A segment is a consecutive run of slots assigned to the same skill. A high number of segments per day indicates frequent task switching.

Category 5 — General Scheduling Statistics

These statistics provide a broader view of schedule quality and are particularly useful for benchmarking and guiding heuristic search algorithms. They are grouped into two sub-groups.

5.1 Group A — Workload utilisation

ID	Name and description
STAT-A1	**Mean workforce utilisation** Average contracted hours utilisation across all employees. A high mean close to 100% is desirable. Should be read together with STAT-A2.
STAT-A2	**Variance of workforce utilisation** How unequally working hours are distributed among employees. A high variance means some employees are heavily loaded while others are underused.
STAT-A3	**Max-min workload gap** Difference in total worked hours between the most and least loaded employee. A large gap signals imbalanced scheduling.
STAT-A4	**Fairness index** Normalised Gini coefficient of worked hours across all employees. Ranges from 0 to 1. Values below 0.1 are considered fair.
STAT-A5	**Idle employees** Number of employees whose utilisation falls below 50% of their contracted hours. These employees represent underused capacity.
STAT-A6	**Overloaded employees** Number of employees assigned more hours than their contract allows.
STAT-A7	**Number of assignments used** Total number of daily shift assignments activated in the solution. Can be expressed as a percentage of all possible assignments to give a selection rate.

5.2 Group B — Skill allocation analysis

ID	Name and description
STAT-B1	**Percentage of time allocated to each skill** Share of total worked time devoted to each skill (ALM, EG, CAJ) across all employees. Shows the overall task distribution in the schedule.
STAT-B2	**Mean skill changes per day** Average number of times an employee switches from one skill to another within a single day. Frequent switching may reduce operational continuity.
STAT-B3	**Mean segment duration** Average duration of a continuous skill block (segment) across all employees and days. Longer segments indicate more stable task allocation throughout the day.